WORKED SOLUTION

STEP III 2024, Question 5

Pure • Matrix Differential Equations

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Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 7.7, *Quantities That Never Change*: https://beyondstep.keithmcnulty.org/ch07/lax-pairs.html

The key idea

Everything hangs on two facts about the trace: it is linear, and $\operatorname{tr}(\mathbf{XY}) = \operatorname{tr}(\mathbf{YX})$. Part (ii) is a formula for the rate of change of the determinant (Jacobi's formula), and it turns "is $\det \mathbf{M}$ constant?" into "is a certain trace zero?". In (iii) a commutator $\mathbf{MN} - \mathbf{NM}$ always has trace zero, which gives all three invariants. In (iv) the same formula shows that $\operatorname{tr} \mathbf{N}$ measures how fast $\ln|\det \mathbf{M}|$ changes, so the question becomes whether a matrix can have constant trace but varying determinant.

Solution

Part (i)

Write $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\mathbf{N} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$. The diagonal entries of $\mathbf{MN}$ are $ap + br$ and $cq + ds$, while those of $\mathbf{NM}$ are $pa + qc$ and $rb + sd$. So

$$\operatorname{tr}(\mathbf{MN}) = ap + br + cq + ds = \operatorname{tr}(\mathbf{NM}).$$

Also $\mathbf{M} + \mathbf{N}$ has diagonal entries $a + p$ and $d + s$, so

$$\operatorname{tr}(\mathbf{M} + \mathbf{N}) = \operatorname{tr}(\mathbf{M}) + \operatorname{tr}(\mathbf{N}).$$

In the same way $\operatorname{tr}(k\mathbf{M}) = k \operatorname{tr}(\mathbf{M})$ for a scalar k, and the trace of $\mathrm{d}\mathbf{M}/\mathrm{d}t$ is the derivative of $\operatorname{tr} \mathbf{M}$. We use these freely below.

Part (ii)

With entries a, b, c, d depending on t (dashes denote $\mathrm{d}/\mathrm{d}t$), the product rule gives

$$\frac{\mathrm{d}}{\mathrm{d}t}(\det \mathbf{M}) = \frac{\mathrm{d}}{\mathrm{d}t}(ad - bc) = a'd + ad' - b'c - bc'.$$

Since $\mathbf{M}$ is non-singular, $\mathbf{M}^{-1} = \dfrac{1}{\det \mathbf{M}} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$, and

$$\mathbf{M}^{-1}\frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t} = \frac{1}{\det \mathbf{M}} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} = \frac{1}{\det \mathbf{M}} \begin{pmatrix} da' - bc' & * \\ * & -cb' + ad' \end{pmatrix}.$$

Its trace is $\dfrac{a'd + ad' - b'c - bc'}{\det \mathbf{M}}$, which is $\dfrac{1}{\det \mathbf{M}} \dfrac{\mathrm{d}}{\mathrm{d}t}(\det \mathbf{M})$, as required.

Part (iii)

Determinant. By (ii) and linearity of the trace,

$$\frac{1}{\det \mathbf{M}} \frac{\mathrm{d}}{\mathrm{d}t}(\det \mathbf{M}) = \operatorname{tr}\big(\mathbf{M}^{-1}(\mathbf{MN} - \mathbf{NM})\big) = \operatorname{tr}(\mathbf{N}) - \operatorname{tr}(\mathbf{M}^{-1}\mathbf{NM}).$$

Using (i) with the two matrices $\mathbf{M}^{-1}\mathbf{N}$ and $\mathbf{M}$, we get $\operatorname{tr}(\mathbf{M}^{-1}\mathbf{N}\,\mathbf{M}) = \operatorname{tr}(\mathbf{M}\,\mathbf{M}^{-1}\mathbf{N}) = \operatorname{tr}(\mathbf{N})$. So the right-hand side is zero. Since $\det \mathbf{M} \neq 0$, this gives $\frac{\mathrm{d}}{\mathrm{d}t}(\det \mathbf{M}) = 0$.

Trace. $\dfrac{\mathrm{d}}{\mathrm{d}t} \operatorname{tr}(\mathbf{M}) = \operatorname{tr}\Big(\dfrac{\mathrm{d}\mathbf{M}}{\mathrm{d}t}\Big) = \operatorname{tr}(\mathbf{MN}) - \operatorname{tr}(\mathbf{NM}) = 0$ by (i).

Trace of the square. The product rule holds for matrices provided we keep the order of the factors:

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{M}^2) &= \frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t}\mathbf{M} + \mathbf{M}\frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t} = (\mathbf{MN} - \mathbf{NM})\mathbf{M} + \mathbf{M}(\mathbf{MN} - \mathbf{NM}) \\ &= \mathbf{M}^2\mathbf{N} - \mathbf{NM}^2, \end{aligned}$$

because the two $\mathbf{MNM}$ terms cancel. This has the same commutator form, now with $\mathbf{M}^2$ in place of $\mathbf{M}$, so its trace is zero by (i). Hence $\operatorname{tr}(\mathbf{M}^2)$ is constant too. All three quantities are independent of t.

> **Ask yourself** How can we solve $\mathrm{d}\mathbf{M}/\mathrm{d}t = \mathbf{MN} - \mathbf{NM}$ when $\mathbf{N}$ depends on t as well?

Split off the part of $\mathbf{N}$ that commutes with everything. Write $\mathbf{N} = t\mathbf{I} + t\mathbf{J}$ with $\mathbf{J} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. The identity commutes with $\mathbf{M}$, so $\mathbf{MN} - \mathbf{NM} = t(\mathbf{MJ} - \mathbf{JM})$. With $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$\mathbf{MJ} - \mathbf{JM} = \begin{pmatrix} 0 & a \\ 0 & c \end{pmatrix} - \begin{pmatrix} c & d \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} -c & a - d \\ 0 & c \end{pmatrix}.$$

Comparing entries gives four scalar equations:

$$a' = -tc, \qquad b' = t(a-d), \qquad c' = 0, \qquad d' = tc.$$

We solve them in order, using the values A, B, C, D at $t = 0$:

- $c' = 0$, so $c = C$.
- Then $a' = -Ct$ and $d' = Ct$, so $a = A - \frac{1}{2}Ct^2$ and $d = D + \frac{1}{2}Ct^2$.
- Now $a - d = A - D - Ct^2$, so $b' = (A-D)t - Ct^3$, giving $b = B + \frac{1}{2}(A-D)t^2 - \frac{1}{4}Ct^4$.

Therefore

$$\mathbf{M} = \begin{pmatrix} A - \frac{1}{2}Ct^2 & B + \frac{1}{2}(A-D)t^2 - \frac{1}{4}Ct^4 \\ C & D + \frac{1}{2}Ct^2 \end{pmatrix}.$$

As a check, the trace is $A + D$, and expanding $ad - bc$ shows that every power of t cancels, leaving $AD - BC$. This agrees with the first half of the part.

Part (iv)

Now $\mathrm{d}\mathbf{M}/\mathrm{d}t = \mathbf{MN}$, so $\mathbf{M}^{-1}\,\mathrm{d}\mathbf{M}/\mathrm{d}t = \mathbf{N}$, and (ii) gives

$$\operatorname{tr}(\mathbf{N}) = \frac{1}{\det \mathbf{M}} \frac{\mathrm{d}}{\mathrm{d}t}(\det \mathbf{M}).$$

So $\operatorname{tr}\mathbf{N} = 0$ for all t exactly when $\det\mathbf{M}$ is constant. A constant trace says nothing about the determinant, so the answer should be *no*. To prove it we need one counterexample.

> **Ask yourself** How do we build a counterexample, when the question gives us $\mathbf{N}$ only through the differential equation?

Work backwards. Choose any non-singular $\mathbf{M}(t)$ with constant non-zero trace and non-constant determinant, and *define* $\mathbf{N} = \mathbf{M}^{-1}\,\mathrm{d}\mathbf{M}/\mathrm{d}t$. Then the equation holds automatically. Take

$$\mathbf{M} = \begin{pmatrix} 1 & t \\ -t & 1 \end{pmatrix}, \qquad \operatorname{tr}\mathbf{M} = 2, \qquad \det\mathbf{M} = 1 + t^2 \neq 0.$$

Then

$$\mathbf{N} = \mathbf{M}^{-1}\frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t} = \frac{1}{1+t^2}\begin{pmatrix} 1 & -t \\ t & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \frac{1}{1+t^2}\begin{pmatrix} t & 1 \\ -1 & t \end{pmatrix}.$$

Multiplying out confirms that $\mathbf{MN} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \mathrm{d}\mathbf{M}/\mathrm{d}t$. Also $\mathbf{N}$ is non-singular, since $\det\mathbf{N} = 1/(1+t^2)$. Yet

$$\operatorname{tr}(\mathbf{N}) = \frac{2t}{1+t^2},$$

which is non-zero for every $t \neq 0$. So it is **not** necessarily true that $\operatorname{tr}(\mathbf{N}) = 0$. What *is* forced is $\operatorname{tr}(\mathbf{MN}) = \operatorname{tr}(\mathrm{d}\mathbf{M}/\mathrm{d}t) = 0$, which is not the same thing.

Checks

- In (iii), take $A = 1$, $B = 0$, $C = 2$, $D = 3$ and $t = 1$. The formula gives $\mathbf{M} = \begin{pmatrix} 0 & -\frac{3}{2} \\ 2 & 4 \end{pmatrix}$, which has trace 4 and determinant 3, the same as at $t = 0$. Its square has trace $0 - 3 - 3 + 16 = 10$, and so does the initial matrix: $1 + 0 + 0 + 9 = 10$.
- The formula for $\mathbf{M}$ in (iii) agreed to eight decimal places with a computer integration of the differential equation for random A, B, C, D. The three invariants stayed constant when the same integration was run with a random time-dependent $\mathbf{N}$.
- In (iv), at $t = 1$ we have $\mathbf{N} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ with trace 1, while $\det \mathbf{M} = 1 + t^2$ has logarithmic derivative $2t/(1 + t^2) = 1$ there, as (ii) predicts.

Common slips

- Writing $\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{M}^2) = 2\mathbf{M}\frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t}$: matrices need not commute, so both orders must be kept.
- Assuming $\operatorname{tr}(\mathbf{MN}) = \operatorname{tr}\mathbf{M} \operatorname{tr}\mathbf{N}$. That is false. Only the linear rule and the cyclic rule hold.
- In (iv), arguing from $\operatorname{tr}(\mathbf{MN}) = 0$ straight to $\operatorname{tr}(\mathbf{N}) = 0$. A "no" answer needs an explicit counterexample.

Beyond STEP

Part (ii) is the 2×2 case of Jacobi's formula, which holds for $n \times n$ matrices. Combined with $\mathbf{M}' = \mathbf{MN}$ it gives $\det \mathbf{M}(t) = \det \mathbf{M}(0) \exp \int_0^t \operatorname{tr}\mathbf{N}$. For $\mathbf{N}$ constant this becomes the famous identity $\det e^{\mathbf{N}} = e^{\operatorname{tr}\mathbf{N}}$. Equations of the form $\mathbf{M}' = \mathbf{MN} - \mathbf{NM}$ are called Lax equations. The quantities $\operatorname{tr}(\mathbf{M}^k)$ are always conserved, which is why they appear in the theory of integrable systems. Strang's *Introduction to Linear Algebra* is a good place to read about traces, determinants and matrix exponentials.

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